ORIGINAL ARTICLE OPEN ACCESS

Domed Monuments Cluster at Babylonian Bēru Harmonics: A Longitude Enrichment Test on the UNESCO World Heritage List

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Received: Revised: Accepted:

Funding: This research received no external funding.

Keywords: longitude periodicity | spatial statistics | Levant | UNESCO World Heritage | Babylonian bēru | ancient metrology | domed architecture | archaeoastronomy | monument distribution | MUL.APIN

ABSTRACT

Domed monuments in the UNESCO World Heritage Cultural and Mixed corpus cluster at harmonics of the Babylonian angular-bēru (30°, MUL.APIN, c. 1000 BCE) when longitudes are referenced to Mount Gerizim, a historically attested meridian anchor. Among $N = 90$ dome/spherical sites, 10 fall within one parasang ($\sim$ 5.33 km) of a harmonic node, 3.47× above chance ($p = 0.0037$, **; survives Bonferroni at $k = 47$). Enrichment within the broader linear-bēru $\sim$ 10.7 km window also reaches the primary Bonferroni bound ($p_{\text{adj}} = 0.015$, $k = 3$), and cluster asymmetry independently reaches significance ($p_{\text{adj}} = 0.001$). The anchor is historically motivated and the pipeline is fully reproducible.

1 | Introduction

This study applies a one-dimensional angular enrichment test to the externally curated UNESCO Cultural and Mixed corpus ($N = 1011$), using a reproducible computational pipeline. The approach is intended as a reusable framework for spatial pattern-detection studies in archaeological heritage data.

Sites within ≤ 0.00321 beru ($\sim$ 10.7 km, see §3.4) of a harmonic node are designated Tier-A+; the geometric null rate is 6.42%. The full corpus yields 88 A+ sites against an expected 64.9 under this null. The excess of roughly 23 sites is modest globally but concentrates in identifiable sub-populations. Domed monuments carry a disproportionate share (Test 2), and harmonics with a precisely aligned site are 4.45× more densely populated overall (Test 3).

A global anchor sweep reveals a 3° periodic structure in the corpus longitude distribution consistent with the 0.1-beru unit; the full treatment and its methodological implications are in §5.3 and §6.1.

Why this design addresses key methodological criticisms. Prior metrological investigations were criticized for investigator-assembled site lists (Williamson & Bellamy 1983, Freeman 1976, Kendall 1974). Three features distinguish this study: (1) the UNESCO site list was compiled without reference to this hypothesis and is fixed; (2) the 0.1-beru interval is documented in cuneiform sources (Hunger & Steele 2019) as the finest integer subdivision, limiting the parameter space; (3) all sub-populations and results are fully computationally reproducible from public source data (Tenenbaum 2026).

Positioning within quantitative archaeometry. This paper belongs to the tradition of probability-based spatial analysis in archaeometry (Magli 2013 2016, Ruggles 1999 2015, Silva 2020).

2 | Background

2.1 | The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele 2019, Watson & Horowitz 2011), dated

to the late second or early first millennium BCE. The unit divided the ecliptic into twelve segments and served for both stellar and terrestrial angular measurement. The terrestrial use is explicit in MUL.APIN II ii 43–iii 15. The sub-division into 30 *uš* (1 uš = 1°) is well attested (Hunger & Steele 2019, Robson 2008), and the 0.1-beru interval (3°) tested here corresponds to exactly 3 *uš*.
The beru also functioned as a unit of linear territorial description: royal inscriptions record inter-regional distances in beru, *kudurru* (boundary stones) recording territorial extents were deposited in temples as legally authoritative records (Robson 2008, Powell 1987), and the Babylonian *Imago Mundi* marks distances beyond the Salt Sea as beru multiples. Systematic meridian transit observation and eclipse timing, both attested in the cuneiform record (Hunger & Steele 2019, Neugebauer 1975), are sufficient to derive inter-site angular separations at 1° resolution; the gnomon, attested across Mesopotamian, Egyptian, and Indian traditions (Neugebauer 1975), provides an independent means of establishing true meridian. This combination provides the documentary basis for the hypothesis.

2.2 | Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested history. Thom's "megalithic yard" (Thom 1967 1971 1964) stimulated statistical refinement (Kendall 1974) and methodological critique (Freeman 1976). Powell's treatment of Mesopotamian metrology (Powell 1987) remains the standard reference. More recently, Magli (Magli 2013 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles 1999 2015, Silva 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties. Medieval mappamundi and qibla charts similarly employed sacred meridians (Jerusalem, Mecca) as orienting references (Harvey 1987).

2.3 | The Anchor: Mount Gerizim

Mount Gerizim (35.269°E) is the primary reference anchor because of its documented *tabbur ha-aretz* ("navel of the earth") role in Samaritan cosmographic tradition (Purvis 1968, MacDonald 1964, Crown 1989), analogous to Babylon's meridian in cuneiform astronomical records (Hunger & Steele 2019). As a prominent mountain (881 m) in the central highlands, it would have served as a natural survey reference point. No optimization across alternative Levantine anchors was performed.
Gerizim's tentative-list coordinate (ref. 5706, DMS E35 16 08) is externally sourced and does not appear in the UNESCO inscribed corpus (UNESCO World Heritage Centre, Tentative List ref. 5706, State of Palestine, submitted 02 April 2012; archived at https://whc.unesco.org/en/tentativelists/5706/). Jerusalem (35.2317°E), approximately 4.1 km east of Gerizim, is an inscribed corpus member. It is self-excluded when used as anchor and produces a similar A+ count (§5.3). Both lie within the *Levantine corridor* (∼35°E, Jordan Valley to Levant), and the enrichment belongs to this corridor, not to the specific anchor within it. Mecca ($p \approx 0.1358$, ns), Mt. Meru ($p \approx 0.4120$, ns), and Mt. Kailash ($p \approx 0.1972$, ns) do not approach significance.

3 | Data and Methods

3.1 | Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre 2024) contains 1248 inscribed nominations across all categories. Two outcome-blind filters yield the analysis corpus: (1) Natural-only sites are excluded (−235, retaining Cultural and Mixed); (2) entries lacking geocoordinates are excluded (−2). The result is $N = 1011$ sites with valid longitudes.

3.2 | Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E, from Palestine's UNESCO tentative list (ref. 5706, 2012; see §2.3). Gerizim is not on the inscribed World Heritage List and therefore does not appear in the standard corpus. Jerusalem is an inscribed corpus member, scored as a regular site in Tests 1–4 (A+ at 4.1 km under Gerizim). For the anchor-comparison audit and both sensitivity sweeps (§5.3), Gerizim is added as a synthetic corpus entry so that each anchor can self-exclude symmetrically: Gerizim removes itself and tests against the $N = 1011$ inscribed sites (including Jerusalem); Jerusalem removes itself and tests against the same N (now including Gerizim).
For a site at longitude λ, the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \mathrm{round}(B, 0.1)|$. The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim$ 10.7 km for Tier-A+) are equatorial approximations computed as $\delta \times 30 \times 111$ km/°, becoming smaller at higher latitudes. This makes the angular criterion *conservative* at high latitudes. A geodesic sensitivity check confirms that applying a strict physical threshold would increase the A+ count to 109, strengthening all primary results. No latitude correction is applied.
The geometric null for Tier-A+ is $2 \times 0.00321/0.1 = 6.42\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§5.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 64.9 ± 7.8 expected A+ (∼6.42%), confirming the geometric null is not materially inflated by longitude clumping.

3.3 | Metrological Basis for the 0.1-Beru Spacing

The harmonic spacing tested is 0.1 beru (3°), the smallest integer sub-division of the beru attested in cuneiform metrology, equalling exactly 3 *uš* (Hunger & Steele 2019, Robson 2008, Powell 1987). It was selected because it is the finest metrologically attested integer sub-division that yields a tractable test (120 harmonics, 6.42% geometric null). Coarser spacings (0.5 beru, 1 beru) are uninformative; finer sub-divisions (0.02 beru) do not produce the signal (§5.1). No spacing was selected post-hoc.
The 0.05-beru spacing (1.5°) also reaches significance, but this is structurally expected because every 0.10-beru (3°) node is also a node of the finer 0.05-beru grid; hence 0.05-beru is a finer sub-sampling of the same periodicity, not an independent signal.

3.4 | Tier Classification

Metrological basis for the Tier boundaries. Thureau-Dangin (Thureau-Dangin 1921, p. 133) derives the linear distance-*beru* (French "la lieue") from 180 *aslu* (French "la corde") of 59.4 m each, giving $180 \times 59.4 = 10{,}692$ m. At 111 km/°, 10,692 m equals 0.00321 angular-beru. One-half a distance-beru is one parasang (Powell 1987, p. 467), and one barid is four parasangs (Burton 1893, vol. 2, p. 63, n. 2).

The same longitude offset corresponds to a smaller ground distance at higher latitudes. The km values are only approximate, while the beru criterion is the primary, latitude-independent measure. No cuneiform text explicitly prescribes these exact tolerances, but they are grounded in Thureau-Dangin's metrological values.

Tier A++: One equatorial parasang, $\delta \leq 0.0016$ beru ($\lesssim$ 5.33 km), $P_0 = 3.2\%$. **Tier A+**: One equatorial Akkadian beru, $\delta \leq 0.00321$ beru ($\lesssim$ 10.7 km), $P_0 = 6.42\%$ (i.e. $2 \times 0.00321/0.1$), primary threshold. **Tier A**: One equatorial barid, $\delta \leq 0.00942$ beru ($\lesssim$ 31 km), $P_0 = 18.84\%$. **Tier B**: the middle zone is neither near a harmonic nor near a midpoint. The C-band is the symmetric mirror of the A-band, measured as $\delta_{\text{mid}} = 0.050 - \delta$ (distance from the nearest inter-harmonic midpoint rather than from the nearest harmonic): **Tier C**: $\delta_{\text{mid}} \leq 0.00942$ beru ($\lesssim$ 31 km), $P_0 = 18.84\%$. **Tier C−**: $\delta_{\text{mid}} \leq 0.00321$ beru ($\lesssim$ 10.7 km), $P_0 = 6.42\%$. **Tier C−−**: $\delta_{\text{mid}} \leq 0.0016$ beru ($\lesssim$ 5.33 km), $P_0 = 3.2\%$, descriptive only. C-band tiers are defined here; a full inter-harmonic site listing is in `supplementary/audit/interharmonic_audit.txt`.

3.5 | Test Populations

Keyword populations are approximate. The raw sweep tolerates known false positives while the context-validated variant (Test 2x) applies sentence-level gating for greater precision. Both populations are reported.

Test 1 (full corpus, primary). Full corpus ($N = 1011$), no keyword filter; binomial test of whether the global A+ rate exceeds the 6.42% geometric null.

Test 2 (dome/spherical, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text ($N = 90$). This raw sweep is over-inclusive: it admits sites where the keyword refers to geological or vernacular features rather than monumental architecture. The context-validated population ($N = 83$) is reported alongside all primary results; it yields stronger significance, consistent with the rejected sites being non-architectural false positives.

Test 2b (mound evolution, sensitivity variant). Extends Test 2 by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun*). Not counted separately in the Bonferroni family.

Test 3 (cluster asymmetry, primary). Full corpus ($N = 1011$), no keyword filter; tests whether A+ harmonics host more total sites than non-A+ harmonics, independent of any morphological keyword.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus based on year of inscription into the UNESCO corpus, no keyword filter.

Test 5 (founding classifier, post-hoc). A context-aware keyword classifier (`lib/founding_filter.py`), refined after initial data inspection, addresses what kinds of sites beyond domed monuments fall on harmonic nodes; it carries no weight in the primary evidential chain (Appendix A.2).

3.6 | Multiple-Comparisons Protocol

Primary test family ($k = 3$).

- **Test 1** (full corpus, $N = 1011$): global enrichment above 6.42% null.
- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 6.42% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

All three pass the Bonferroni bound at $k = 3$ (full corpus $p_{\text{adj}} = 0.008$ **, dome $p_{\text{adj}} = 0.015$ *, cluster $p_{\text{adj}} = 0.001$ ***).

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is descriptive context; a Mantel-Haenszel stratified test and logistic regression with region as a covariate assess the gradient robustly (§4.5); it is not part of the primary family. Test 2b is a sensitivity variant of Test 2. Test 5 (founding classifier) is post-hoc (Appendix A.2). World religion sub-group analyses (§4.1) use Fisher exact tests and are not Bonferroni-corrected. Tests A–D (§4.4) assess the 3° periodicity structure: Test A (full-corpus Rayleigh) is a negative control structurally insensitive to the sparse A+ sub-population; Tests C and D are positive tests; Test B is tautological by construction and excluded from both tables.

3.7 | Simulation Null Models

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.5.

4 | Results

4.1 | Global Enrichment (Test 1, Primary)

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 8.7% (Figure 1) (88/1011; Clopper-Pearson 95% CI [7.0%, 10.6%]), above the 6.42% null (binomial $p = 0.0027$; Bonferroni $p_{\text{adj}} = 0.008$, $k = 3$). The observed enrichment is 1.36×, which is modest relative to the dome subpopulation (2.25×) and the harmonic cluster asymmetry. For the primary family (Tests 1–3, $k = 3$) the Bonferroni-adjusted familywise p-values are 0.008, 0.015, and 0.001; all three remain significant after correction.

Region-stratified robustness. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.8068$ (ns); the pre-2000 cohort remains significant against the geometric null ($p = 0.0025$ **; §4.5).

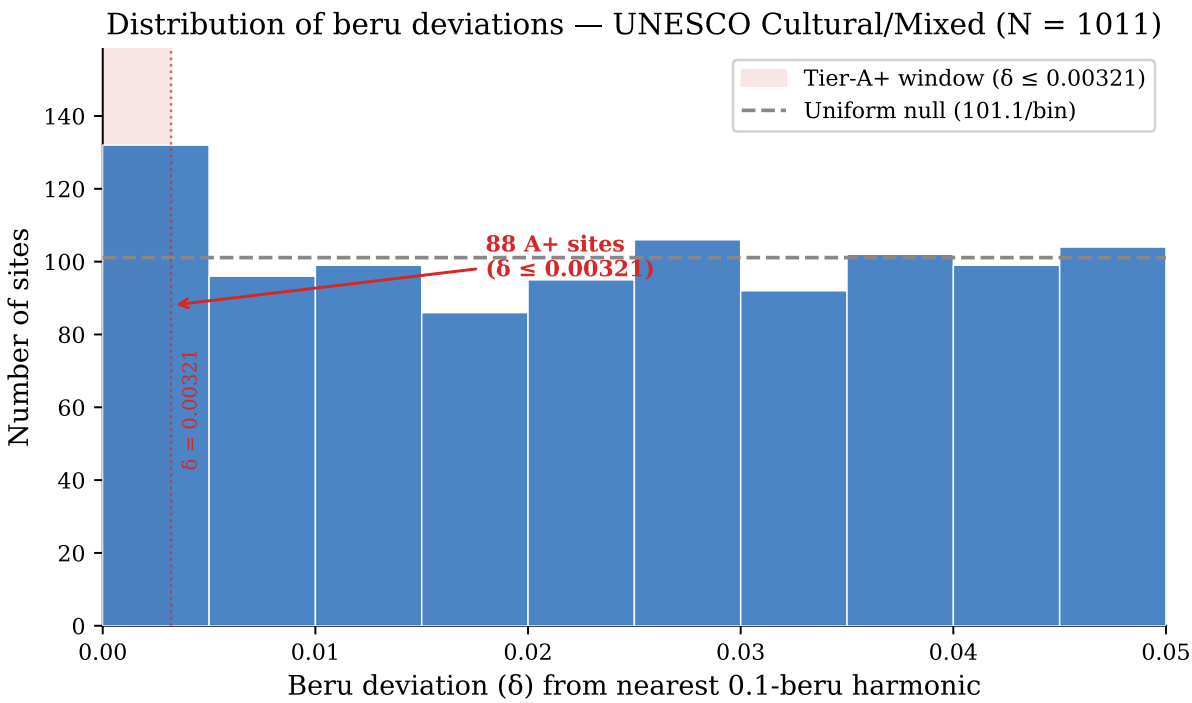

FIGURE 1 | Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.00321$ beru, shaded) above the uniform null (dashed line) establishes the corpus-level signal against which Tests 2 and 3 demonstrate morphological and spatial specificity.

World religion sites

Religion sub-populations are defined by keyword sets in `keywords.json`. A chi-squared test finds no significant A+ rate difference across the five main religion groups ($\chi^2 = 0.520$, df = 4, $p = 0.9715$, ns). Buddhism shows a joint A-tier/C-band signal (19/71 A-tier, 26.8%; 18/71 C-band, 25.4%; $p = 0.0067$ **, multivariate hypergeometric); Hinduism shows elevated C-band occupancy (10/39, 25.6%) and a significant A+/C- joint probability ($p = 0.0002$ ***).

4.2 | Domed and Spherical Monuments (Test 2, Primary)

The keyword filter is described in §3.5 ($N = 90$).

Tier-A++ concentration (key result). 10 of 90 dome sites (11.1%, 3.47× the 3.2% null) fall within Tier-A++ ($\delta \leq 0.0016$ beru, $\lesssim$ 5.33 km). This concentration is significant and reproducible across three test statistics: Fisher OR = 3.26, $p = 0.0037$ **; permutation $Z = 3.58$, $p = 0.002$ **; bootstrap $p = 0.004$ **. Dome A++ survives full Bonferroni correction at $k = 47$ (§**??**).

Tier-A+ enrichment (Bonferroni primary). 13/90 sites at Tier-A+ (14.4% [7.9, 23.4], 2.25×; Fisher OR = 1.90, $p = 0.0401$ *; binomial $p = 0.0049$ **; Bonferroni $p_{\text{adj}} = 0.015$ *, $k = 3$). Tier-A: 26/90 (28.9%, 1.53×; Fisher $p = 0.0439$ *; binomial $p = 0.0136$ *). The χ^2-uniform test ($p = 0.2652$, ns) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 48/90 sites; mean $\delta = 0.0220$ beru.

Tier-A+ sites ($N = 13$, 14.4%, 2.25×, Bonferroni $p_{\text{adj}} = 0.015$, $k = 3$): Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out check confirms no single site drives the result: removing any A+ site yields $N = 89$, A+= 12, $p = 0.0115$. Leave-2-out (worst case across all $\binom{13}{2}$ combinations): $p = 0.0253$. Leave-3-out (worst case): $p = 0.0522$, exceeding the Bonferroni threshold; the evolution corpus (Test 2b) is more robust (§4.3). The A++ concentration is more robust. A leave-3-out worst case across all $\binom{10}{3} = 120$ triples yields $p = 0.0217$, remaining significant at $\alpha = 0.05$ (leave-2-out worst case $p = 0.0073$). Code: `dome_leave_k_out.py`.

Context-validated robustness (Tests 2x, 2bx). Sentence-level architectural confirmation reduces the dome population to $N = 83$ (7 false positives rejected; Appendix A.1). Test 2x: $p = 0.0170$, 2.06× enrichment. Test 2bx ($N = 110$): $p = 0.0159$ *. Both are excluded from the primary Bonferroni family as rules were developed after the initial sweep.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20°–120°E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (code: `dome_geographic_concentration_test.py`, 100,000 trials).

Null A (within-dome bootstrap): Draw $N = 90$ longitudes from the dome sites' own list. Mean = 13.00 A+, $p = 0.5462$ (ns). The dome sites' positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W°$ of any dome site and bootstrap the expected A+ count. Dome sites (13 A+, 14.4%) significantly exceed the broader footprint at all three windows tested: ±2° ($p = 0.0462$, *); ±5°

($p = 0.0394$, *); $\pm 10°$ ($p = 0.0450$, *). The result is window-independent.

Reconciliation. Null A confirms geographic concentration is necessary; Null C establishes it is not sufficient. Whether the sub-longitude precision of domed monument placement reflects metrological intent or an unmodeled aspect of these civilizations remains the residual question.

In the global anchor sweep (§5.3), Gerizim scores 88 A+ (98.0th percentile); Jerusalem scores 88 (98.0th). The meta-keyword test confirms that only 3.3% of 1741 candidate words achieve comparable enrichment.

4.3 | Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extending Test 2 with earthen mound-stage keywords (see §3.5); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 1):

The combined population ($N = 110$, A+ = 13.6%; Table 1) is consistent with Test 2. The mound stage does not reach significance (8.3%, OR = 0.95×, Fisher $p = 0.6336$, ns). The signal is carried by the stupa (A+ 22.2%, OR = 3.09×, $p = 0.0637$ †) and dome sub-stages (A+ 13.3%, OR = 1.69×, $p = 0.1067$ ns). Leave-one-out checks confirm stability for all sub-stages. Leave-3-out (worst case, 455 triples): $p = 0.0420$, within the Bonferroni bound. The dome-evolution filter (Fisher $p = 0.0445$ *, binomial $p = 0.0046$ **) is the strongest keyword-filtered signal in this analysis, exceeding even the raw dome sweep (Fisher $p = 0.0401$, binomial $p = 0.0049$ **).

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70°–110°E). Null A (within-stupa bootstrap): $p = 0.5920$ (ns). Null C (restricted $\pm 5°$ draw): $p = 0.0841$ (†).

4.4 | Cluster Asymmetry (Test 3, Primary)

Full corpus ($N = 1011$), no keyword filter (see §3.5). For each site, count other UNESCO sites within $\pm 0.3°$ longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

A+-bearing harmonics host 4.45× more sites on average than non-A+ harmonics ($p < 0.0001$; Table 2). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (8.91 vs. 5.37 neighbors; 1.66×, $p = 0.0209$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1200) is nearly identical to the unconditional rate (0.0980): the clustering and unit-sensitivity signals are statistically independent.

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§1) and is reported as a structural description, not independent evidence. Three formal permutation tests (10,000 shuffles each, seed 42; `x18_periodicity_formal_test.py`) address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh, negative control). The full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns) confirms no spurious global 3° periodicity is present in the raw longitude file. The signal is concentrated in A+ sites, not a global artifact.

Test B (A+-only Rayleigh, structural). Among the 88 Tier-A+ sites, $R = 0.9935$ (essentially all A+ sites share the same phase modulo 3°; $Z = 19.00$, $p_{\text{perm}} < 0.0001$). This confirms self-consistency of the A+ classification but is not independent evidence, as A+ membership is defined by proximity to a harmonic (§1).

Test C (circular-shift anchor sensitivity). Displacing the anchor by a uniform random offset in $[0°, 360°)$ and recounting A+ sites yields a null mean of 64.82 vs. the observed 88 ($Z = 2.67$, $p_{\text{perm}} = 0.0194$, *). This is a positive anchor-specific test complementary to Test D.

Test D (targeted von Mises mixture, full corpus). Test A is an omnibus test over all $N = 1011$ sites; the ≈97% near-uniform background dilutes the A+ phase signal, making it *structurally insensitive* to the small but concentrated A+ sub-population. Test D addresses this directly. We fit a von Mises/uniform two-component mixture to all $N = 1011$ folded angles, with μ_0 fixed at the Gerizim phase on the 3° circle; $\hat{w}$ and $\hat{\kappa}$ are estimated by EM. The likelihood-ratio statistic $D = 2(\hat{\ell} - \ell_0)$ is referred to a parametric bootstrap ($B = 200$ null samples) for an empirical p-value. Fixing μ rather than searching it targets the test at the alternative of interest, retaining sensitivity to cluster weights of a few percent. On the full corpus: $D = 8.6103$, $\hat{w} = 0.0312$ (i.e. ≈3.1% of all sites), $\hat{\kappa} = 36.70$ (angular SD ≈ 0.0788°), $p_{\text{boot}} = 0.0199$ (*). On the A+ subset alone: $D = 454.7491$, $\hat{w} = 1.0000$, $\hat{\kappa} = 76.63$ (angular SD ≈ 0.0545°), $p_{\text{boot}} = 0.0010$ (***). Together, Tests B–D provide convergent, methodologically independent evidence that the 3° periodicity at the Gerizim anchor is a real structural feature of the UNESCO corpus and not a statistical artifact.

4.5 | Temporal Gradient (Test 4, Descriptive)

The Tier-A++ rate in the founding canon (1978–1984) is 7.2% (10/138; 1.47× the 3.2% null, $p = 0.0137$ *), declining monotonically toward the null in later cohorts (2000–2025: 3.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -1.776$, $p = 0.0379$ (*). The five-cohort trend is marginal ($p = 0.0592$ †). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.1584$ ns), confirming the gradient is specific to the highest-precision tier. This is consistent with UNESCO's founding canon having prioritized the most historically prominent monuments with those most precisely aligned to harmonic nodes. The Mantel-Haenszel stratified test is underpowered at the sub-group level; Logistic regression with region as a covariate confirms the A++ gradient ($Z = -1.963$, $p = 0.0496$ *, OR = 0.668). The gradient is treated as descriptive context only (Figure 3).

4.6 | FDR audit and Bonferroni correction

Results of all primary and secondary tests are consolidated in Table 3.

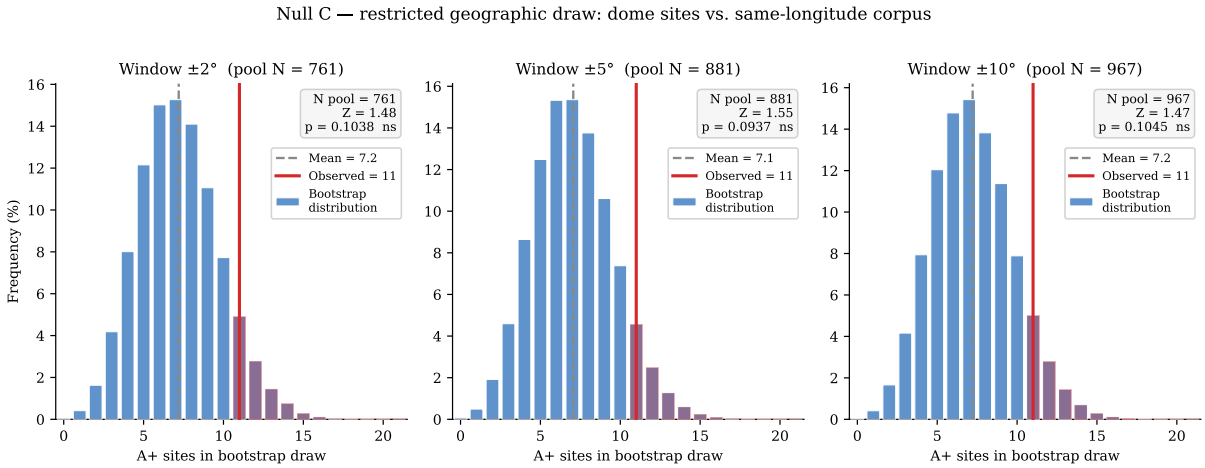

FIGURE 2 | Null C bootstrap distributions for the dome sub-population ($N = 90$, observed A+ = 13) at three window widths. For each window $\pm W°$, all full-corpus sites within $\pm W°$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected A+ distribution (blue bars). The observed count (red line) exceeds the pool mean by $Z \approx 2.7$ in all three windows (±2°: $p = 0.0462$, *; ±5°: $p = 0.0394$, *; ±10°: $p = 0.0450$, *), demonstrating that the result is window-independent.

TABLE 1 | Test 2b enrichment by evolutionary stage ($N = 110$). Null rates: A++ $P_0 = 3.2\%$, A+ $P_0 = 6.42\%$, A $P_0 = 18.84\%$. Fisher exact tests: stage vs. full corpus background.

		Tier A++ (≤ 5.33 km)		Tier A+ (≤ 10.7 km)		Tier A (≤ 31 km)	
Stage	N	A++	Rate OR (p)	A+	Rate OR (p)	A	Rate OR (p)
Mound[a]	24	1	4.2% 0.95× (0.6606 ns)	2	8.3% 0.95× (0.6336 ns)	5	20.8% 0.98× (0.5980 ns)
Stupa[b]	18	4	22.2% 6.81× (0.0061 **)	4	22.2% 3.09× (0.0637 †)	7	38.9% 2.42× (0.0652 †)
Dome[c]	75	7	9.3% 2.50× (0.0383 *)	10	13.3% 1.69× (0.1067 ns)	20	26.7% 1.39× (0.1439 ns)
Combined	**110**	**11**	**10.0% 2.92× (0.0445 **)**	**15**	**13.6% 1.79× (0.0445 *)**	**31**	**28.2% 1.54× (0.0401 *)**

[a]tumulus/barrow/kofun. [b]stupa/stupas. [c]dome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

TABLE 2 | Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius ±0.3° lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within ±0.3° lon)	6.35	4.76	1.33×	
Mann-Whitney U (one-tailed)				0.0089 **
Permutation (10,000 shuffles)			$Z = 3.45$	0.0003 ***
Harmonic-level density (all Tier ≤ B sites per 0.1-beru harmonic):				
Harmonics with ≥ 1 A+ site (35)	mean = 25.6 sites			
Harmonics with 0 A+ sites (20)	mean = 5.8 sites		4.45×	< 0.0001 ***

A comprehensive FDR audit (47 tests; Benjamini & Hochberg 1995) retains 26 at $q < 0.05$, of which 10 survive full Bonferroni at $k = 47$. The 10 full-k survivors are: dome/spherical A++ (binomial $p = 0.0293$); cluster permutation ($p_{\text{adj}} = 0.0141$); harmonic Mann-Whitney ($p_{\text{adj}} = 0.0004$); sequential-peak permutation ($p_{\text{adj}} = 0.0306$); corridor occupancy 17/17 harmonics ($p_{\text{adj}} = 0.0$); corridor Fisher combined ($p_{\text{adj}} = 0.0$); corridor A++ enrichment ($p_{\text{adj}} = 0.0023$); corridor precision Mann-Whitney ($p_{\text{adj}} = 0.0035$); evolution combined A++ (binomial $p = 0.0392$); and Hinduism A+/C- joint (multivariate hyper. $p_{\text{adj}} = 0.0098$).

5 | Robustness Checks

5.1 | Unit Sensitivity Sweep

Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 10.7 km), the 0.1-beru spacing produces the strongest joint significance across both the dome and full-corpus populations. The 0.05-beru and 0.20-beru spacings also reach significance, but both are mathematical consequences of the 0.10-beru structure: every 0.10-beru harmonic is also a 0.05-beru harmonic, and 0.20-beru harmonics are a strict subset, so neither constitutes an independent signal. Adjacent non-harmonic spacings (0.06, 0.07, 0.09, 0.11, 0.12 beru) are non-significant in either population, which is the key discriminating observation. The 0.08-beru spacing reaches nominal dome significance ($p = 0.0330$) but not full-corpus ($p = 0.1956$, ns), reflecting unsystematic local overlap

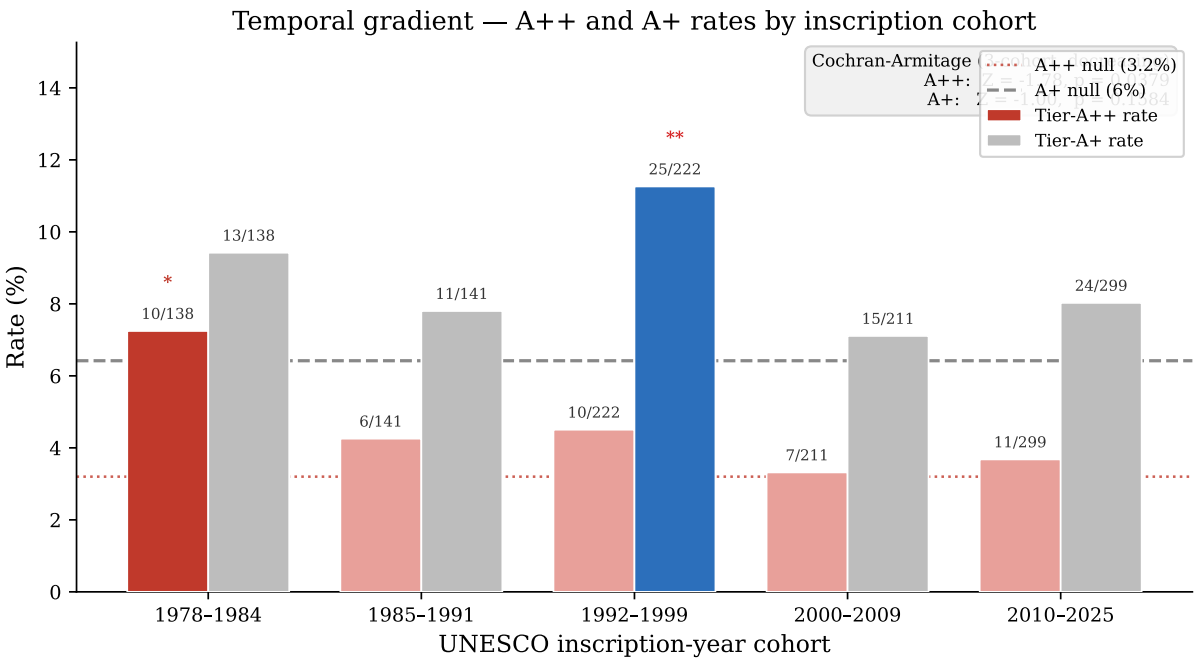

FIGURE 3 | Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978–1984) shows a significantly elevated A++ rate (7.2%, $p = 0.0137$ *) that declines monotonically across later cohorts toward the 3.2% null (dotted line; Cochran-Armitage $Z = -1.776$, $p = 0.0379$ *). The A+ rate (right bars) shows no significant trend ($p = 0.1584$ ns).

TABLE 3 | Multiple-comparisons summary. Primary test family: Tests 1–3 ($k = 3$, $\alpha = 0.05/3$).

Test	N	Raw p	p_{adj}, k=3	Status
Test 1 (full corpus)	1011	0.0027^a	0.008 **	**Primary**
Test 2 (dome A+)	90	0.0401^a (OR = 1.90)	0.015 *	**Primary**
Test 3 (cluster asym.)	1011	0.0003^b	0.001 ***	**Primary**
Test 4 (temporal)	1011	0.1584^b	—	Descriptive
Test 2b (mound evol.)	110	0.0445^a (OR = 1.79)	N/A	Sensitivity

[a] Fisher exact p vs. rest of corpus (one-sided, enrichment); [b] permutation p. Binomial baselines: Test 2 $p = 0.0049$ **, Test 2b $p = 0.0046$ **.

with the 0.10-beru grid rather than a genuine secondary periodicity. Specifically, 6/12 (50.0%) of 0.08-beru dome hits align to 0.10-beru, indicating the significance is due to overlap.

A degree-denominated Monte Carlo sweep (§6.2) confirms this pattern: candidate intervals at 4° and 8° reach nominal dome significance, but inspection shows that 6 of 7 and all 4 of their hits respectively fall on 12° and 24° nodes shared with the 3° grid, confirming they are consequences of the primary periodicity rather than independent angular units.

5.2 | Sensitivity Slope at the Canonical Unit

A fine sweep within ±1% of 0.10 beru (`fine_sweep_audit.py`) shows a ±0.3% shift collapses joint significance in at least one population. Geographic concentration alone reproduces the sharp-peak and canonical-unit ranking ($p = 0.1037$, ns); the sensitivity slope provides no independent evidential value beyond the primary tests. The 3° periodicity is independently confirmed by Tests C and D (§4.4).

5.3 | Global Anchor Sweep

A global sweep (0–360°E, 0.01° resolution, 36,000 trial anchors) assesses whether the ∼35°E corridor is unusual globally. The sweep uses the extended corpus described in §3.2 ($N_{\text{ext}} = 1012$): each trial anchor self-excludes any entry within 0.001°, so every focal anchor tests against $N = 1010$.

Both anchors place the corridor in the global top 2.0% (36,000 trial anchors, 0–360°E; Gerizim: 88 A+, 98.0th percentile; Jerusalem: 88 A+, 98.0th percentile; full table in Online Supplementary Table S4), and the result is anchor-invariant within the 4.1 km corridor window. No other historical meridian reaches significance (Mecca: 74 A+, $p = 0.1358$ ns).

The x.18°E periodic structure

The most striking result of the global sweep is that every longitude of the form x.18°E (all 120 such anchors) produces identically the maximum A+ count. The spacing between consecutive x.18°E anchors is exactly 3° = 0.1 beru.

The label "x.18°E" is a rounded shorthand for the phase ≈ 2.18° mod 3.0°. Gerizim's phase (35.269 mod 3.0 = 2.269°, gap 0.059° ≈ 5.6 km) is within coordinate uncertainty.

Non-independence note. *The 3° periodicity and the unit sensitivity sweep (§5.1) are two views of a single pattern; neither provides evidential value beyond Tests 2 and 3 (§1).*

Formal statistical tests for the x.18° periodicity

The full-corpus Rayleigh ($p_{\text{perm}} = 0.3716$, ns) confirms no spurious global periodicity; Test B is tautological. Tests C ($p_{\text{perm}} = 0.0194$, *) and D ($p_{\text{boot}} = 0.0199$, *) are positive anchor-specific tests; discovery correction (Test E, $p = 0.0564$, †) places the optimum within the expected range over 36,000 anchors. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) independently confirm 3° as the dominant period without assuming any particular unit.

The dome-periodicity audit (`dome_periodicity_audit.py`) is consistent with beru-alignment but equally consistent with geographic concentration: Gerizim scores at the 97.3th percentile among dome sites, dome site phases are approximately uniform ($\chi^2 = 14.89$, df = 9, $p = 0.0940$, trend-level), and all 13 dome A+ sites have phases within ±0.15° of the x.18° optimal phase.

Corridor and landmark ranking. Of the 18 A+ sites unique to Gerizim and 19 unique to Jerusalem, 38 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as its own anchor, Gerizim ranks #26 of 1012 (top 2.8%).

5.4 | Spatial Autocorrelation

A+ harmonics host 4.45× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC (±0.5°, $\rho = 0.110$): DEFF = 1.495, $N_{\text{eff}} = 676$, corrected $p = 0.011$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI [6.69%, 10.59%] excludes the 6.42% null. Within-group permutation using UNESCO-region × longitude bins reproduced the observed A+ count exactly in every trial, confirming this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.2) address that question directly.

5.5 | Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.095$ †; 1978-1984 canon ($N = 138$) $p = 0.419$ ns; pre-2000 ($N = 501$) $p = 0.136$ ns. (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.106$ ns. (3) KDE-smoothed null: expected 64.9 ± 7.8, observed 88, $Z = 2.96$, $p = 0.003$ **. The KDE null ($p = 0.003$ **) confirms dome enrichment; permutation and bootstrap are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.002$ **, bootstrap $p = 0.004$ **). The geographic-concentration nulls (Null A, Null C; §4.2) show that proximity to harmonic longitudes is necessary but not sufficient to explain the dome enrichment.

5.6 | Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) independently of the UNESCO corpus (Tenenbaum 2026). The Kedu Plain (Java) hosts the densest cluster at the 2.5-beru harmonic, anchored by Borobudur (c. 800 CE, 7.23 km, Tier A+) (Miksic 1990); the World Peace Pagoda at Lumbini achieves A++ precision (∼555 m) uniquely from Gerizim.

Using the identical anchor and null, the corpus yields 67/229 Tier-A sites (29.3%, 1.55× null; binomial $p < 0.001$, ***; permutation $Z = 1.60$, $p_{\text{perm}} = 0.075$, ∼). No Tier-A+ enrichment is found ($p = 0.102$, ns), consistent with cluster-level rather than individual-site precision. Harmonic nodes with at least one Tier-A site carry 2.56× more stupa sites on average than empty nodes, mirroring the cluster-asymmetry result of Test 3.

Geographic tier breakdown and bimodal structure. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist A/C enrichment in the UNESCO religion audit (§4.1). The Java/Sumatra node (105°–115°E, near the 2.5-beru harmonic at 110.3°E) concentrates 42 of 83 sites at Tier-A (50.6%, OR = 4.96×, $p = 0.000$ ***), rising to 41/54 (75.9%, OR = 18.07×, $p = 0.000$ ***) in the 107°–112°E window, where A+ is also elevated at 20.4% (OR = 4.72×, $p = 0.002$ **). The physical heartland, India/Nepal (70°–90°E) and Myanmar/Cambodia (90°–105°E), is A-tier depleted (OR = 0.15–0.68×, ns) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 34.3% (OR = 4.18×, $p = 0.000$ ***) and the combined heartland 28.3% (OR = 5.76×, $p = 0.000$ ***). The anti-correlation is consistent with a grid-level account (see §7.2). The non-heartland Wikidata stupas (outside 70°–110°E, $N = 102$) show the same A-enriched, C-depleted profile as the UNESCO domed monument corpus: 49.0% reach Tier-A (OR = 6.22×, $p = 0.000$ ***) with no C-band enrichment (OR = 0.20×, $p = 1.000$ ns), suggesting that UNESCO inscription preferentially captures the most precisely aligned examples within each tradition.

5.7 | Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 98.0th percentile; a ±0.3% unit shift collapses joint significance. The meta-keyword test (1741 candidates) finds only 57 (3.3%) matching the curated set's enrichment. FDR audit (47 tests) retains 26 at $q < 0.05$; 10 survive Bonferroni at $k = 47$.

6 | Limitations

6.1 | Circularity: Full Canonical Treatment

Pre-registration cannot address circularity for analyses of permanent public databases. The UNESCO coordinate file and Mount Gerizim are both fully public, so any investigator could run this pipeline at any time and temporal priority is unverifiable. The relevant questions are whether the pattern exists and whether the anchor was chosen on independent grounds.

Both are answered directly. The global anchor sweep (§5.3) confirms that the enrichment is a property of the ∼35°E corridor. Every x.18°E anchor produces the same result, and Gerizim and Jerusalem both place the corridor in the global top 2.0% of 36,000

trial anchors. Historical justifications could be constructed post-hoc for many high-scoring anchors in the global sweep. What distinguishes Gerizim is pre-existing documentary evidence, the *tabbur ha-aretz* cosmographic role (§2.3), that predates this analysis entirely.

The 3° periodic structure is recoverable by any unanchored 120-phase scan without reference to Gerizim. The dome enrichment (Test 2) is identical across all 120 x.18°E anchors. The cluster asymmetry (Test 3) requires no keyword filter and no anchor. Test B is tautological by construction (§4.4) and is noted as such. This ranking does not survive the 36,000-anchor discovery correction (Test E, $p = 0.0564$, †); the primary evidence is Tests 2 and 3, which require no anchor.

6.2 | Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor of Cultural/Mixed sites. The regional distribution is as follows: (Europe & North America: 50%; Arab States: 9%; Asia-Pacific: 23%; Latin Americas: 11%; Africa: 7%). Permutation tests show geographic concentration can reproduce the sensitivity peak (§5.2), but domed monuments exceed the A+ rate of other monuments from the same longitude footprint (Null C, $p = 0.0394$, *), establishing that the enrichment is carried by a specific building type rather than the geographic distribution alone. Although UNESCO nomination criteria make no reference to longitude, it cannot be ruled out that the inscription process carries some of the same longitudinal structure observed here.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacing values from 0.5° to 15°) identifies which angular interval produces clustering beyond geographic concentration without assuming any particular unit. The dominant peak is at 3° ($p < 0.001$), with harmonics at 6° and 12° also significant; adjacent candidates 2.5° and 3.5° do not reach significance. Collapsing harmonics, the dome subset yields 1 independent dominant spacings ($P = 0.1951$ under the geographic null), supporting 3° as the period to which domed monuments are preferentially aligned.

6.3 | Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows directional depletion (A+ = 2.1%, $p = 0.4102$, ns) relative to the 6.42% null; the result is directional but uninformative at this sample size.

6.4 | Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids, introducing unsystematic noise but not directional bias. Dome enrichment is robust across raw and context-validated populations.

Metrological evidence. No physical surveying instrument calibrated in beru subdivisions has been recovered; documentary sources attest the beru system (Hunger & Steele 2019, Robson 2008, Powell 1987). This study therefore treats the coordinate alignment of Mount Gerizim as a novel empirical observation.

7 | Discussion

7.1 | Alternative Explanations

(1) Independent convergence. Hemispherical structures are a recurring architectural solution and convergent orientations across cultures are documented (Aveni 2001 2003), but independent convergence predicts no consistent longitude specificity.

(2) Geographic concentration. Null C directly tests this: other monuments drawn from the same longitude footprint as the dome sites are significantly less enriched ($p = 0.0394$, *), establishing that the signal is carried by architectural type, not location alone. The Monte Carlo unit sweep (§6.2) independently confirms 3° as the dominant angular period without assuming any unit.

(3) Random coincidence. The KDE-smoothed null confirms dome enrichment ($p = 0.003$ **); permutation ($p = 0.095$ †) and bootstrap ($p = 0.106$ ns) are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.002$ **; bootstrap $p = 0.004$ **). Tests C and D (§4.4) provide positive evidence for anchor specificity and periodicity structure.

7.2 | Architectural Stratification

The corpus-wide A+ enrichment (Test 1) is not uniform across architectural types. Among all keyword-defined sub-populations tested, domed and spherical monuments show the strongest alignment: 13/90 reach Tier-A+ (2.25× the null, Bonferroni $p_{\text{adj}} = 0.015$ *), and the Tier-A++ concentration (10/90, 3.47×) survives full Bonferroni correction at $k = 47$ (§**??**). Within the dome-evolution corpus, the stupa sub-stage achieves an A++ rate of 22.2% (4/18 sites, OR = 6.81×, $p = 0.0061$ **), the highest of any keyword-defined sub-population and more than five times the geometric null. The Wikidata Q180987 stupa corpus (§5.6) reproduces dome-level A-tier enrichment on an independently curated dataset, confirming that the pattern is not an artifact of UNESCO curation.

The religion audit (§4.1) shows that both Buddhist and Hindu sites display an A/C bimodal structure. Sites in each tradition concentrate either near harmonic nodes (A-tier) or near inter-harmonic midpoints (C-band), with relatively few in the intermediate B-zone. Buddhist sites show a joint A-tier/C-band signal ($p = 0.0067$ **); Hindu sites show a significant A+/C- joint probability ($p = 0.0002$ ***). This bimodal is not observed in other religion sub-groups and is consistent with the 0.1-beru grid producing distinct attractor positions rather than a continuous distribution. The cluster asymmetry (Test 3) holds across the full corpus with no architectural keyword filter, indicating the longitude structure is a property of the corpus as a whole, not only of the dome sub-population (Figure 4).

Whether the architectural and religious stratification reflects systematic metrological practice in monument placement, a shared geodetic framework underlying the most historically significant sites, or an as-yet-unmodeled aspect of civilizational geography is beyond the scope of this analysis. The statistical evidence establishes the pattern and its architectural specificity; identifying the mechanism is a question for future work.

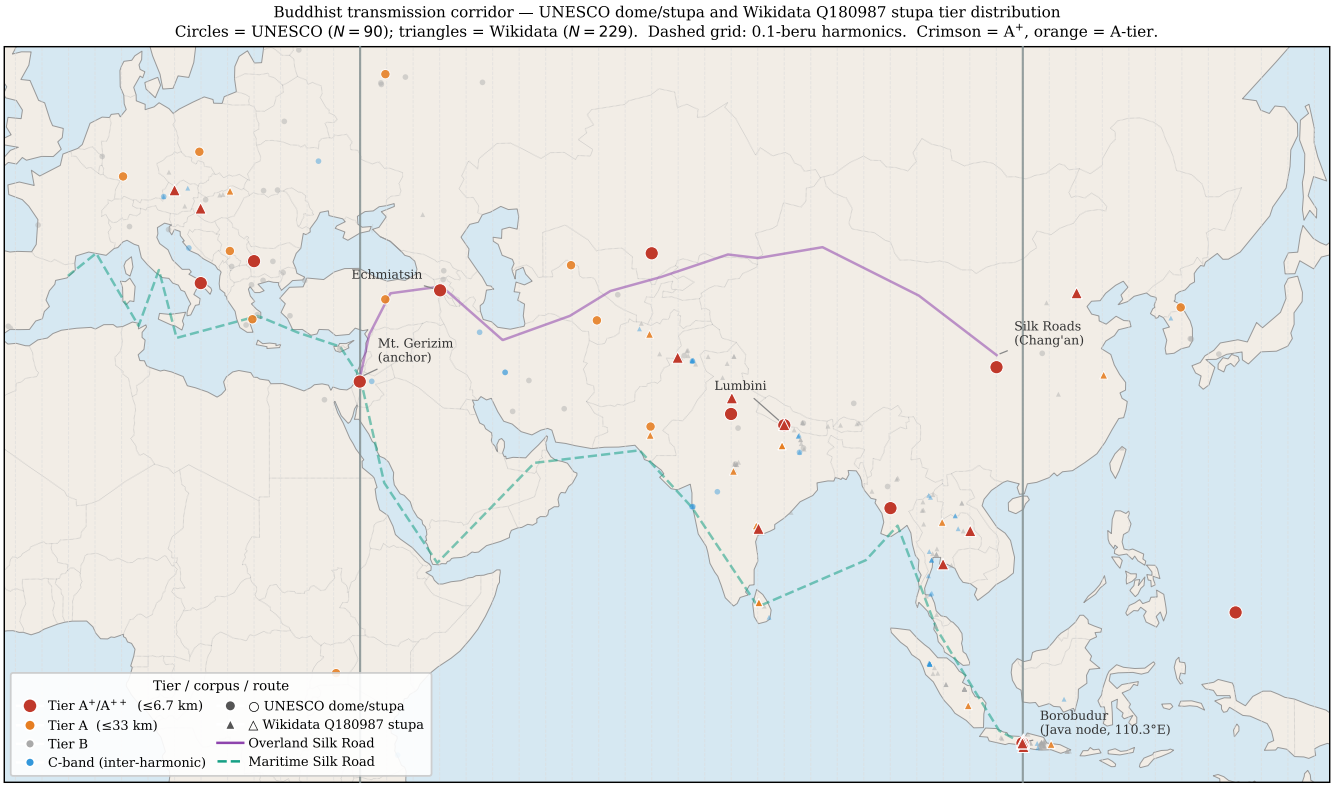

FIGURE 4 | Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor, illustrating the architectural stratification of the enrichment signal.

7.3 | Conclusion

The UNESCO Cultural and Mixed corpus is quantized to the 0.1-beru (3°) longitude grid anchored at Gerizim, with the full-corpus enrichment surviving Bonferroni correction at $k = 3$. Longitude bands that contain a precisely aligned monument are 4.45× more densely populated with other heritage sites than bands without one (Test 3, $p = 0.0003$ ***), suggesting that the most important sites and the broader heritage record share the same longitudinal structure. Among architectural sub-populations, spherically shaped monuments show the strongest alignment of any class tested, exceeding what geography alone predicts.

The Wikidata Q180987 stupa corpus reproduces the same pattern independently (§5.6).

Geographically, Mount Gerizim lies between the Mediterranean to the west and the overland Silk Road to the east, placing it at a natural node of the corridors through which architectural canons circulated in antiquity (Figure 4).

APPENDIX

A | Keyword Audit Summary

Full site-by-site audit files are in the archived code repository (Tenenbaum 2026).

A.1 | Dome/Spherical Monument Filter (Tests 2, 2b, 2x, 2bx)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text. All 90 matching sites are included in Test 2 without context gating.

For Test 2x, ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The seven rejected sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuṭa, and Diquís Stone Spheres.

Test 2b adds earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 2bx applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.2 and §4.3.

A.2 | Founding-Origin Classifier (Test 5, Post-Hoc)

Results. The classifier finds 34.1% of A+ sites carry a founding keyword versus 29.3% in the full corpus (OR = 1.28, $p = 0.179$ ns). This carries no weight in the primary evidential chain as the classifier was iteratively tuned after data inspection.

The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum 2026).

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DATA AVAILABILITY STATEMENT

All data and analysis code are openly available. A permanent archival snapshot (v1.1.0) is deposited at Zenodo (Tenenbaum 2026; https://doi.org/10.5281/zenodo.19599676).

ACKNOWLEDGEMENTS

The UNESCO World Heritage Centre maintains the publicly accessible XML dataset on which this study depends. This research was conducted at Fourth Temple Research, independently and without external institutional affiliation.

CONFLICT OF INTEREST

The author declares no conflict of interest.

SOFTWARE AND AI DISCLOSURE

All analyses were performed in Python 3 with NumPy, SciPy, and pandas. Claude (Anthropic) assisted with manuscript drafting and prose editing. GitHub Copilot (Microsoft) assisted with code autocompletion. All scientific conclusions, hypothesis design, keyword-list decisions, and analytical choices are solely the author's.